THE CLEBSCH CUBIC PROGRAM — IV

A binary $[78, 36, 12]$ code from the passant lines of a conic over $\mathbb{F}_{13}$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, 13)$, and form the binary incidence code on its 78 internal points using the 78 passant lines. We prove that this code has parameters $[78, 36, 12]_2$. It has exactly 364 minimum words, in four $\mathrm{PGL}(2, 13)$-orbits of size 91, and each orbit spans the code. More strongly, the minimum supports reconstruct the passant incidence matrix and the six-class elliptic association scheme; their coordinate-permutation automorphism group is exactly $\mathrm{PGL}(2, 13)$. The proof combines a tangent-product exclusion in weight eight, two exhaustive but compact syndrome-disjointness certificates in weight ten, and structural arguments in the binary association algebra for spanning and reconstruction.

Keywords. binary incidence code; finite conic; passant line; minimum word; association scheme; reconstruction.

1 The passant incidence code

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 13)$. A line is *passant* when it is disjoint from $\mathcal{C}(\mathbb{F}_{13})$. Conic polarity identifies the 78 internal points with the 78 passant lines. Let M be their binary incidence matrix and put

$$K = \ker_{\mathbb{F}_2} M \subseteq \mathbb{F}_2^{78}.$$

The coordinate set is the set of internal points. Automorphisms below are coordinate permutations preserving the indicated code or minimum-support system.

Theorem 1.1 (Minimum-layer reconstruction)**.** *The code K has parameters $[78, 36, 12]_2$ and exactly 364 minimum words. Under $\mathrm{PGL}(2, 13)$, those words form four orbits of size 91: one has stabilizer isomorphic to S_4, and three have dihedral stabilizer of order 24. Every one of the four orbits spans K.*

Pair and triple concurrence in the minimum supports reconstruct the six relations of the elliptic association scheme on the internal points. Among the passant seven-cliques, the 78 with zero triple concurrence are exactly the rows of M. Consequently the minimum-support system reconstructs M up to row order, and its coordinate-permutation automorphism group is exactly $\mathrm{PGL}(2, 13)$.

2 Distance twelve

Row parity gives the elementary lower bound eight. A hypothetical weight-eight word would be an eight-arc saturating every passant pencil. Segre's tangent identity reduces it to a seven-clique in a cyclic 42-vertex compatibility graph; a five-row closure proves that graph has clique number five. In weight ten, pencil parity leaves exactly two local profiles. Separate syndrome-disjointness certificates exhaust both profiles, and an independent dynamic program checks the same empty intersections. An explicit dihedral support supplies a word of weight twelve.

3 Minimum words and the elliptic scheme

Exact orbit representatives generate the four minimum-word orbits and their 364 supports. Pair concurrence distinguishes passant from secant joins, while triple-concurrence profiles distinguish all six elliptic relations. This reduces incidence reconstruction to the intrinsic selection of the 78 zero-triple-concurrence seven-cliques.

4 Spanning and automorphisms

The binary reduction of the elliptic association algebra supplies a short spanning argument: one relation matrix maps into K and is injective on K, while the four orbit Gram matrices are powers of that relation. For automorphisms, three anchors form a regular $\mathrm{PGL}(2, 13)$-orbit and a fourth anchor separates all 78 points by its relation signature.

5 Formalization and exact evidence

The formal development separates general conic/passant-code definitions from the finite $q = 13$ instance. Structural reductions, coverage of the two weight-ten profiles, orbit exhaustion, and the reconstruction interfaces are assigned explicit terminal theorems. The paper's verification manifest distinguishes human arguments, kernel-checked results, compact certificates, and independent trusted replays.

6 Conclusion

The minimum layer determines more than the parameters of the code: it recovers the finite geometry that defined the parity checks. The result therefore gives a concrete reconstruction theorem for a binary incidence code, with the irreducibly finite exclusions isolated from the association- algebra mechanism that explains spanning and symmetry.

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